Jtop tool usage

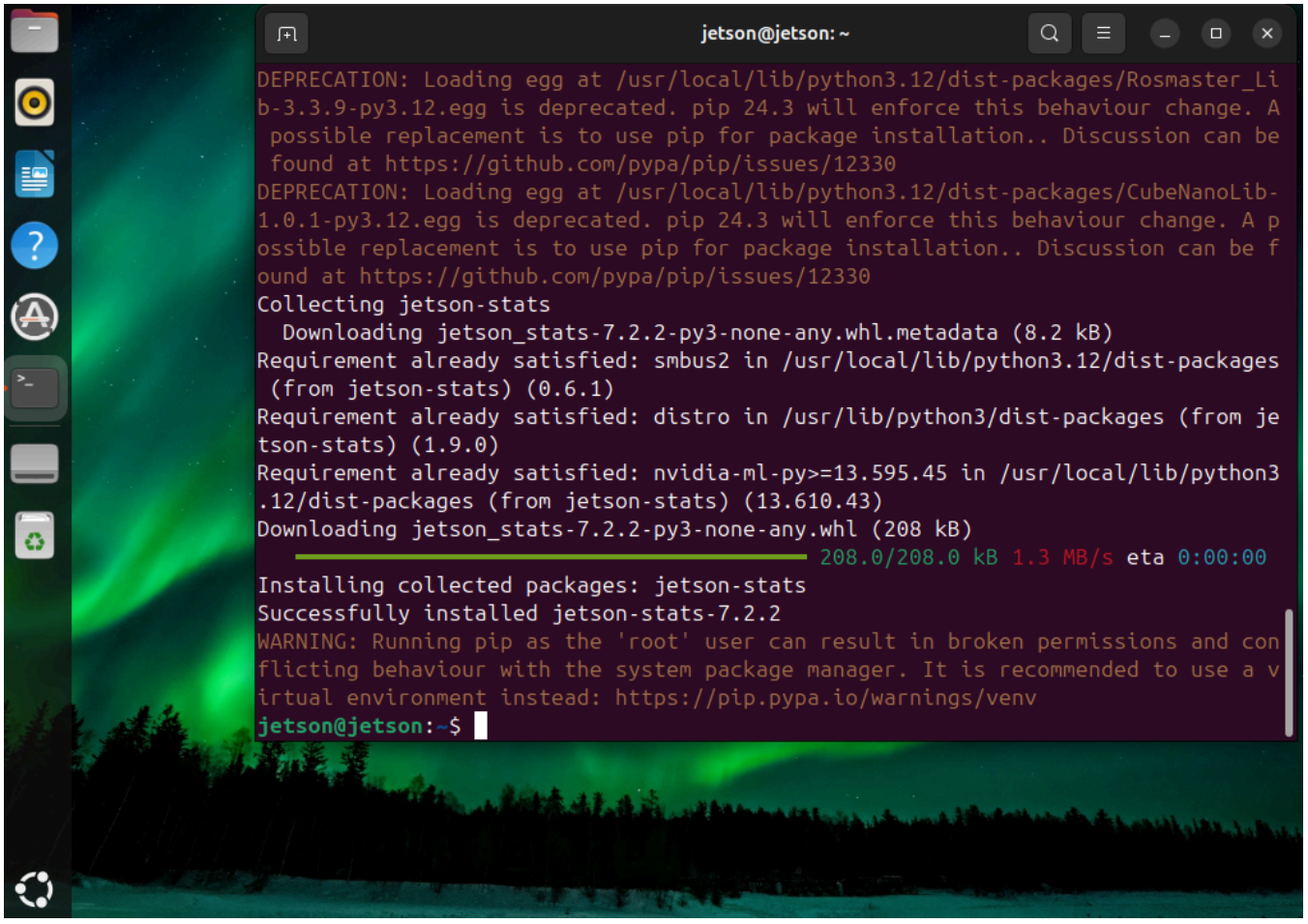
Jtop tool usage

1. Install Jtop
2. Maximum performance mode
    2.1. Enable MAXN mode
    2.2. Enable Jetson clocks
3. Use Jtop

Jtop is a system monitoring tool developed for the NVIDIA Jetson series of devices. It shows the resource usage of every aspect of the Jetson device, such as CPU, GPU, memory, disk and network, and it can display in real time the temperature, power consumption and frequency of the different hardware.

1. Install Jtop

```
sudo apt update
sudo apt install python3-pip -y
sudo pip3 install -U jetson-stats --break-system-packages
```



2. Maximum performance mode

2.1. Enable MAXN mode

Enabling MAXN Power Mode (maximum power mode) on the Jetson ensures that all the CPU and GPU cores are turned on:

```
sudo nvpmodel -m 0
```

2.2. Enable Jetson clocks

Enabling Jetson Clocks ensures that all the CPU and GPU cores run at the maximum frequency:

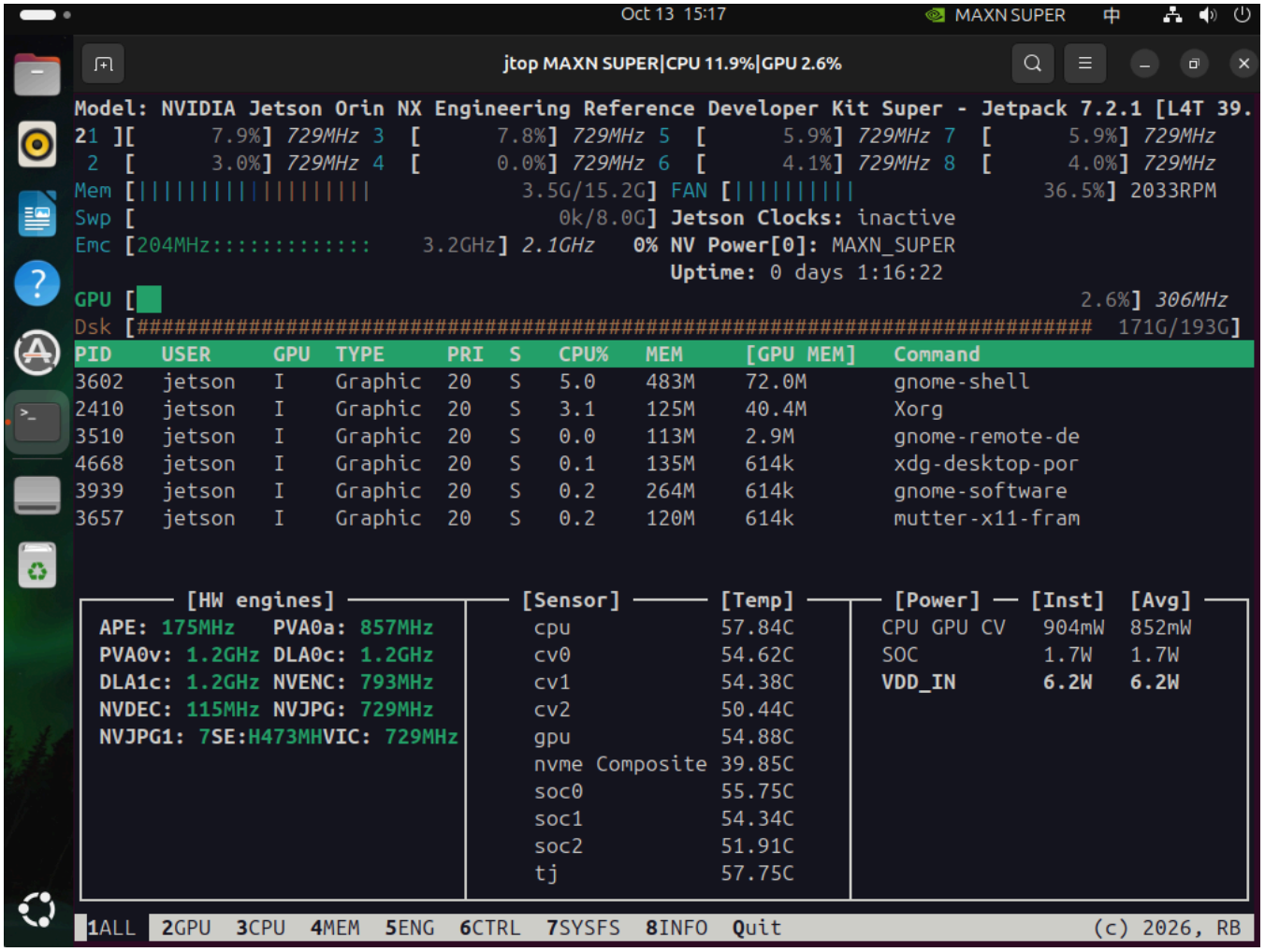
```
sudo jetson_clocks
```

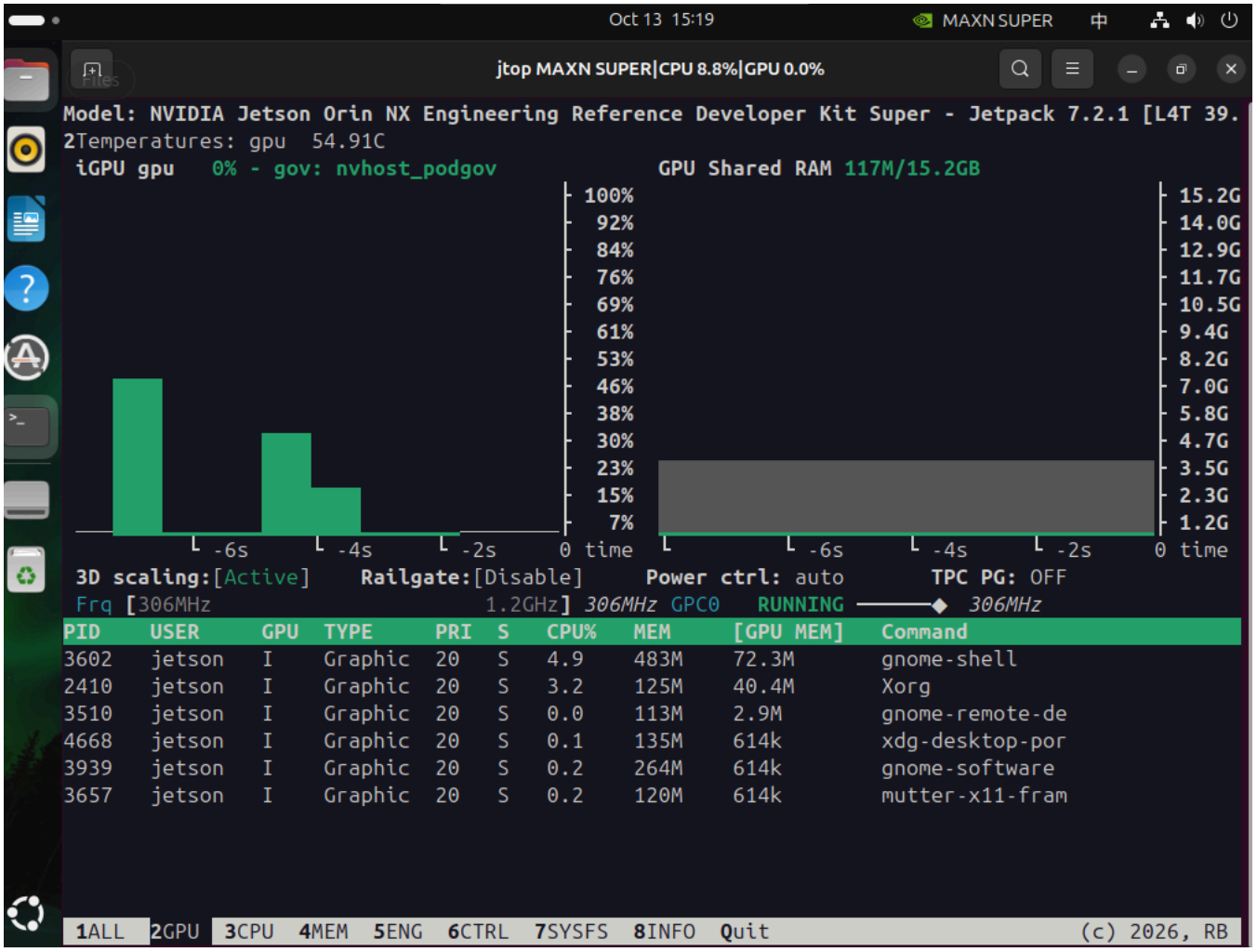
3. Use Jtop

After the system reboots, you can enter the jtop command in the terminal to start the Jtop tool:

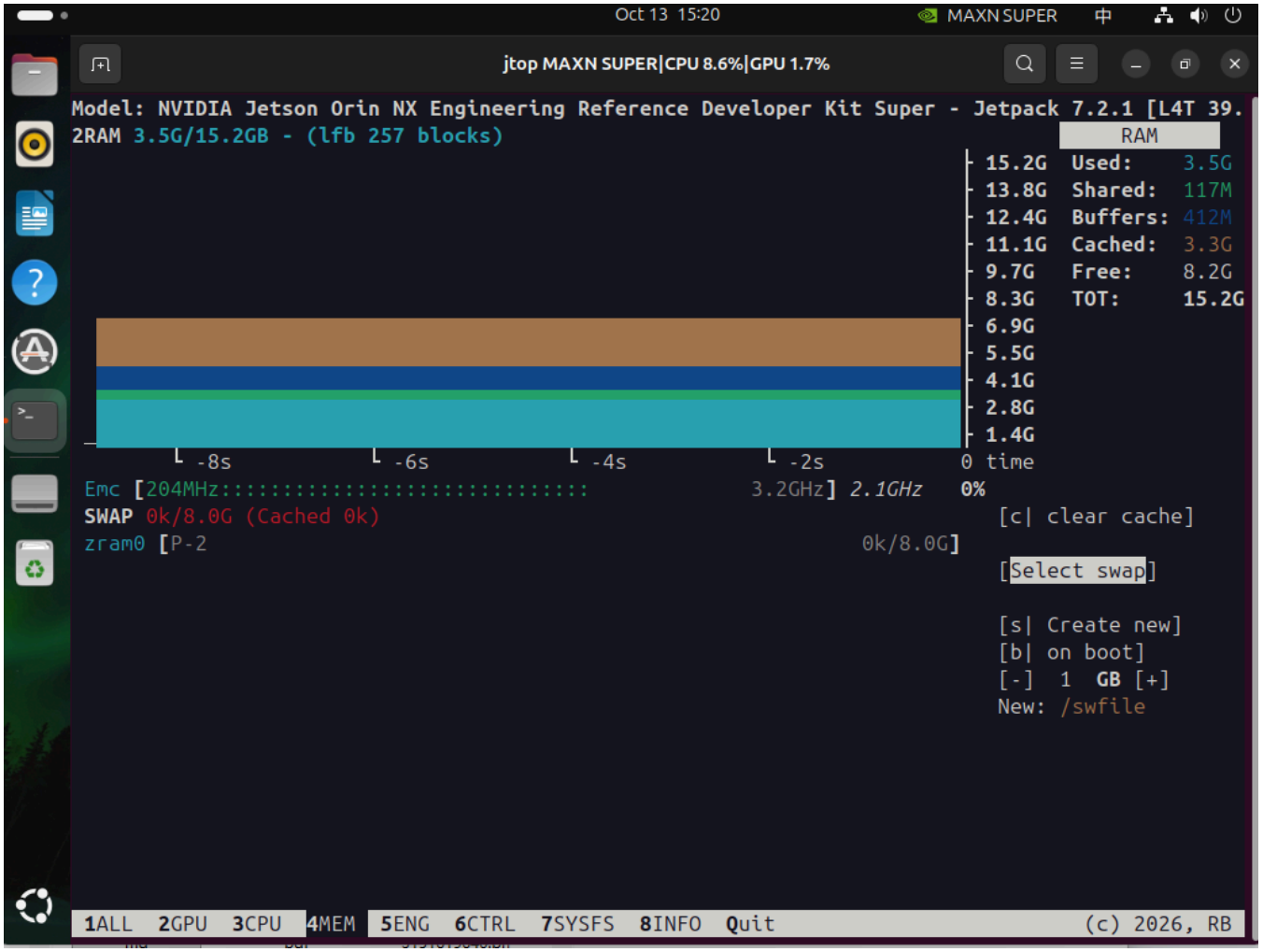
```
jtop
```

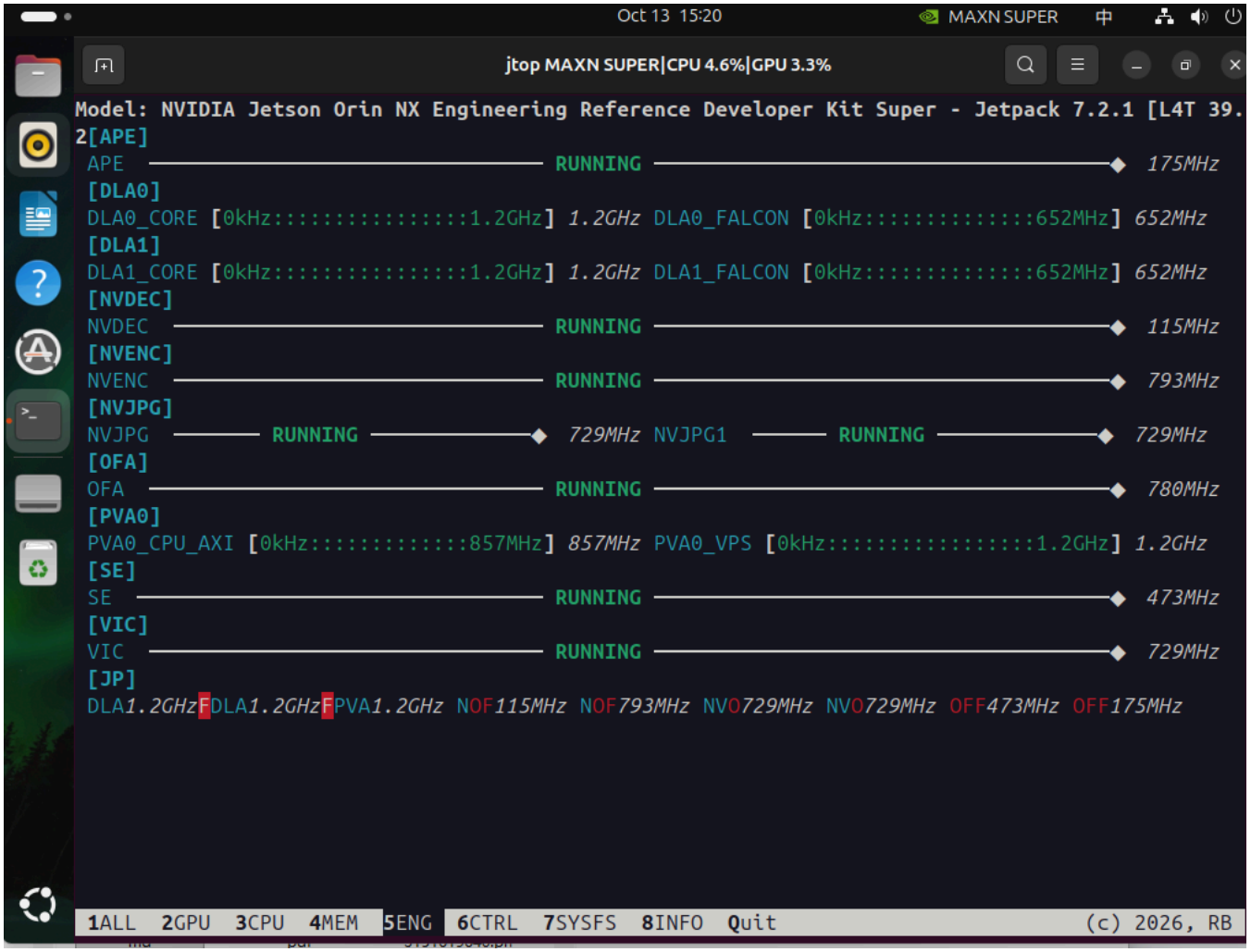
Note: the strongest performance parameters are only displayed when the mainboard power mode is set to MAXN!

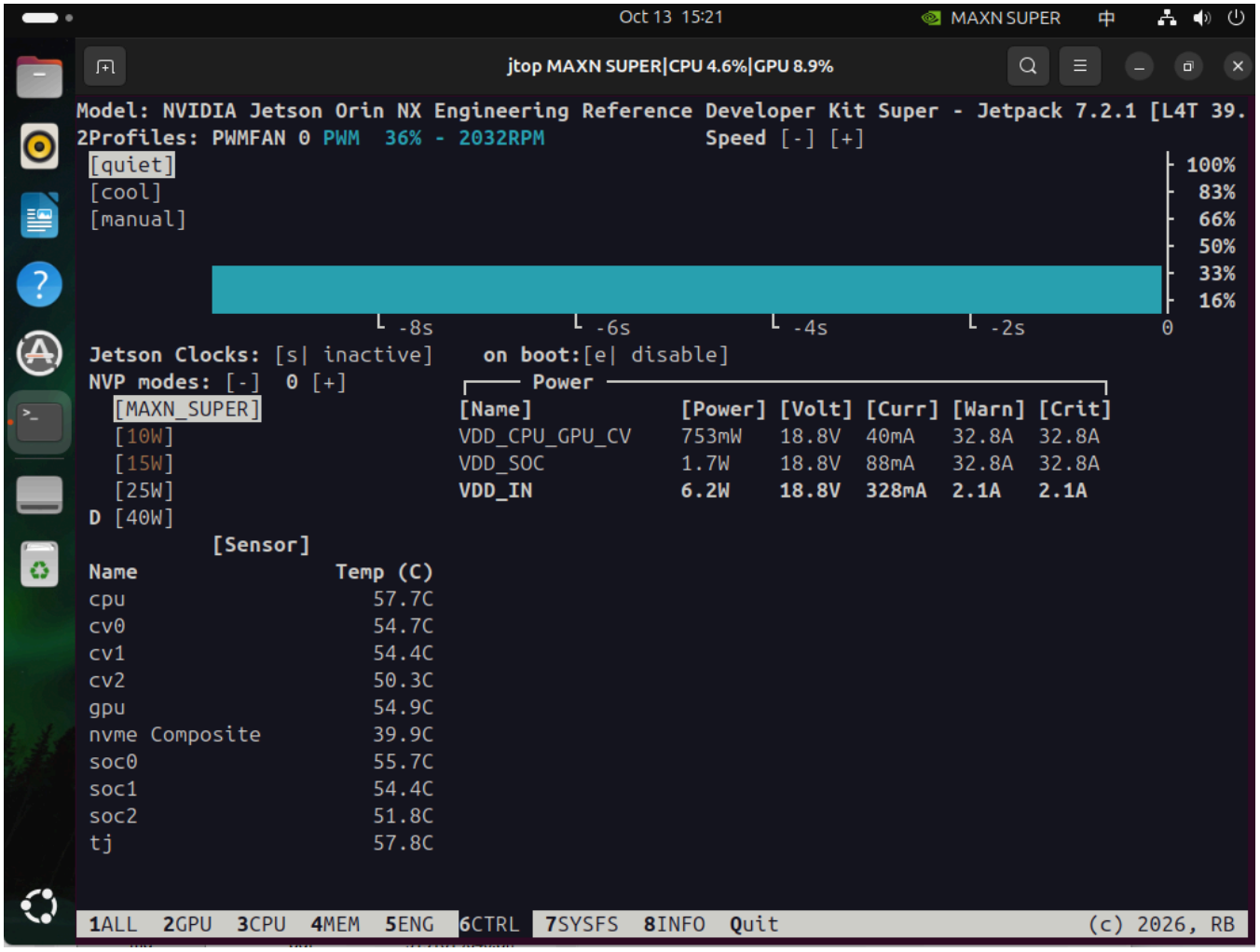


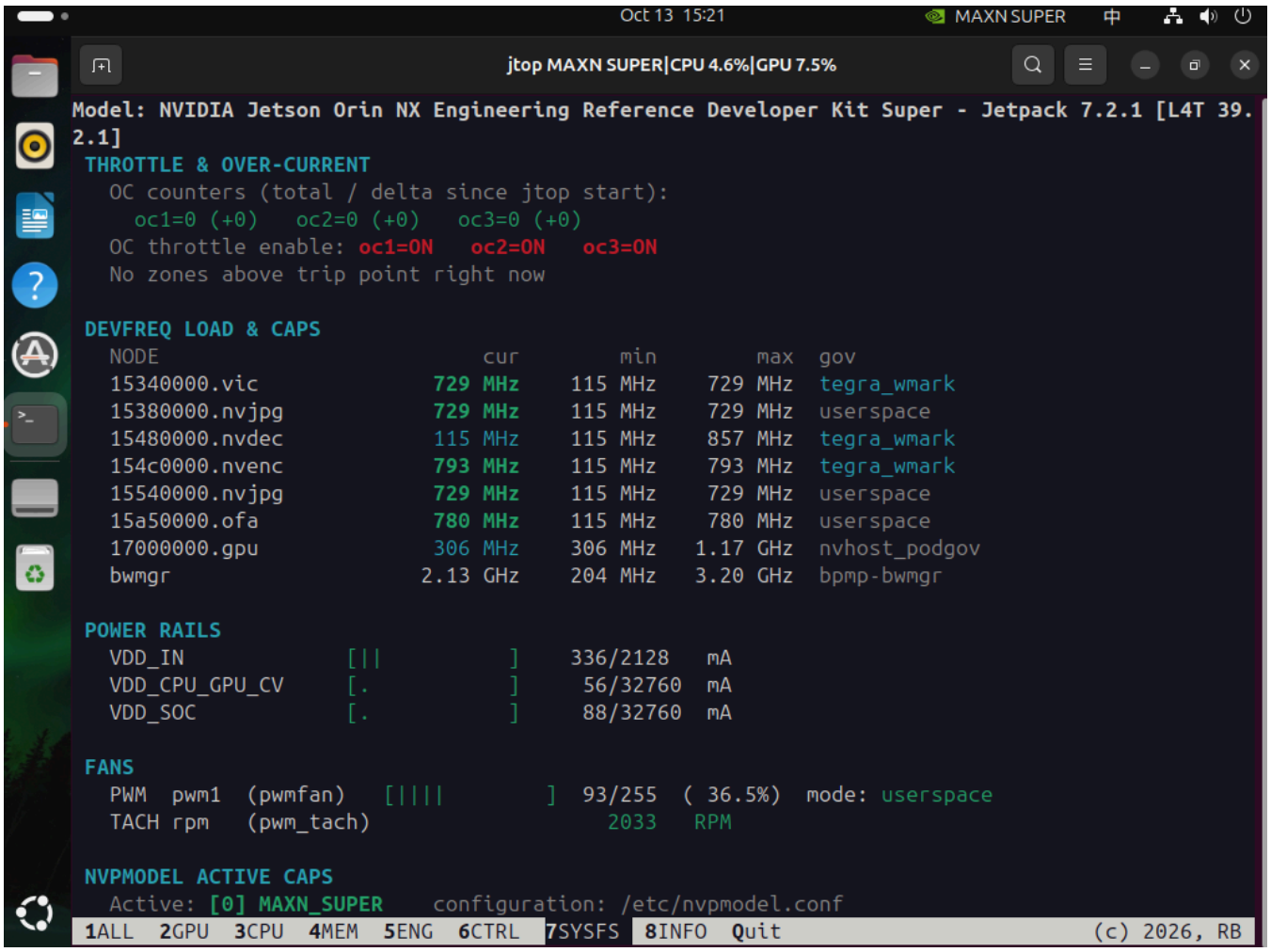

Oct 13 15:19
MAXN SUPER
jtop MAXN SUPER|CPU 8.8%|GPU 0.0%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.
2Temperatures: gpu 54.91C
iGPU gpu 0% - gov: nvhost_podgov
GPU Shared RAM 117M/15.2GB
100%
92%
84%
76%
69%
61%
53%
46%
38%
30%
23%
15%
7%
15.2G
14.0G
12.9G
11.7G
10.5G
9.4G
8.2G
7.0G
5.8G
4.7G
3.5G
2.3G
1.2G
-6s
-4s
-2s
0 time
3D scaling:[Active] Railgate:[Disable] Power ctrl: auto TPC PG: OFF
Frq [306MHz 1.2GHz] 306MHz GPC0 RUNNING 306MHz
PID USER GPU TYPE PRI S CPU% MEM [GPU MEM] Command
3602 jetson I Graphic 20 S 4.9 483M 72.3M gnome-shell
2410 jetson I Graphic 20 S 3.2 125M 40.4M Xorg
3510 jetson I Graphic 20 S 0.0 113M 2.9M gnome-remote-de
4668 jetson I Graphic 20 S 0.1 135M 614k xdg-desktop-por
3939 jetson I Graphic 20 S 0.2 264M 614k gnome-software
3657 jetson I Graphic 20 S 0.2 120M 614k mutter-x11-fram
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit
(c) 2026, RB

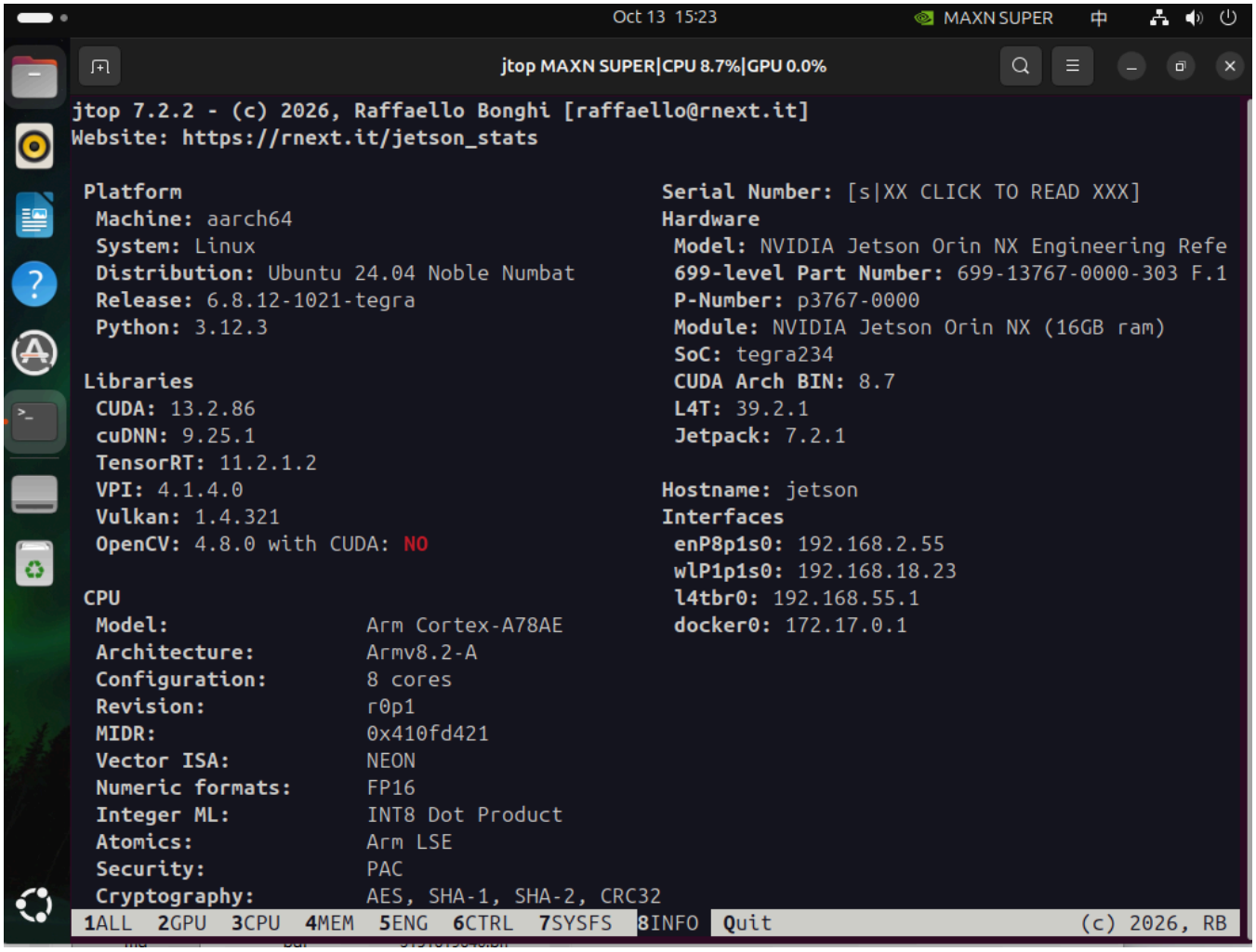

jtop MAXN SUPER|CPU 8.6%|GPU 2.9%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.
2ALL [||||
6.6%]
1 14% Schedutil
4 6% Schedutil
7 4% Schedutil
2 12% Schedutil
5 8% Schedutil
8 9% Schedutil
3 7% Schedutil
6 6% Schedutil
100%
80%
60%
40%
20%
-8s -6s -4s -2s 0
Frq [729MHz 2.0GHz] 729MHz
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit
(c) 2026, RB


Oct 13 15:20
MAXN SUPER
jtop MAXN SUPER|CPU 8.6%|GPU 1.7%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.
2RAM 3.5G/15.2GB - (lfb 257 blocks)
RAM
15.2G
13.8G
12.4G
11.1G
9.7G
8.3G
6.9G
5.5G
4.1G
2.8G
1.4G
Used: 3.5G
Shared: 117M
Buffers: 412M
Cached: 3.3G
Free: 8.2G
TOT: 15.2G
-8s
-6s
-4s
-2s
0 time
Emc [204MHz:: 3.2GHz] 2.1GHz 0%
SWAP 0k/8.0G (Cached 0k)
zram0 [P-2 0k/8.0G]
[c| clear cache]
[Select swap]
[s| Create new]
[b| on boot]
[-] 1 GB [+]
New: /swfile
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit
(c) 2026, RB


Oct 13 15:20
MAXN SUPER
jtop MAXN SUPER|CPU 4.6%|GPU 3.3%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.2[APE]
APE RUNNING 175MHz
[DLA0]
DLA0_CORE [0kHz:::::::::::::::1.2GHz] 1.2GHz DLA0_FALCON [0kHz:::::::::::::652MHz] 652MHz
[DLA1]
DLA1_CORE [0kHz:::::::::::::::1.2GHz] 1.2GHz DLA1_FALCON [0kHz:::::::::::::652MHz] 652MHz
[NVDEC]
NVDEC RUNNING 115MHz
[NVENC]
NVENC RUNNING 793MHz
[NVJPG]
NVJPG RUNNING 729MHz NVJPG1 RUNNING 729MHz
[OFA]
OFA RUNNING 780MHz
[PVA0]
PVA0_CPU_AXI [0kHz:::::::::::::857MHz] 857MHz PVA0_VPS [0kHz:::::::::::::::1.2GHz] 1.2GHz
[SE]
SE RUNNING 473MHz
[VIC]
VIC RUNNING 729MHz
[JP]
DLA1.2GHzFDLA1.2GHzFPVA1.2GHz NOF115MHz NOF793MHz NVO729MHz NVO729MHz OFF473MHz OFF175MHz
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit
(c) 2026, RB


Oct 13 15:21
MAXN SUPER
中
jtop MAXN SUPER|CPU 4.6%|GPU 8.9%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.
2Profiles: PWMFAN 0 PWM 36% - 2032RPM
Speed [-] [+]
[quiet]
[cool]
[manual]
100%
83%
66%
50%
33%
16%
-8s
-6s
-4s
-2s
0
Jetson Clocks: [s| inactive]
on boot:[e| disable]
NVP modes: [-] 0 [+]
[MAXN_SUPER]
[10W]
[15W]
[25W]
D [40W]
Power
[Name] [Power] [Volt] [Curr] [Warn] [Crit]
VDD_CPU_GPU_CV 753mW 18.8V 40mA 32.8A 32.8A
VDD_SOC 1.7W 18.8V 88mA 32.8A 32.8A
VDD_IN 6.2W 18.8V 328mA 2.1A 2.1A
[Sensor]
Name Temp (C)
cpu 57.7C
cv0 54.7C
cv1 54.4C
cv2 50.3C
gpu 54.9C
nvme Composite 39.9C
soc0 55.7C
soc1 54.4C
soc2 51.8C
tj 57.8C
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit
(c) 2026, RB


Oct 13 15:21
MAXN SUPER
jtop MAXN SUPER|CPU 4.6%|GPU 7.5%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.2.1]
THROTTLE & OVER-CURRENT
OC counters (total / delta since jtop start):
oc1=0 (+0) oc2=0 (+0) oc3=0 (+0)
OC throttle enable: oc1=ON oc2=ON oc3=ON
No zones above trip point right now
DEVFREQ LOAD & CAPS
NODE cur min max gov
15340000.vic 729 MHz 115 MHz 729 MHz tegra_wmark
15380000.nvjpg 729 MHz 115 MHz 729 MHz userspace
15480000.nvdec 115 MHz 115 MHz 857 MHz tegra_wmark
154c0000.nvenc 793 MHz 115 MHz 793 MHz tegra_wmark
15540000.nvjpg 729 MHz 115 MHz 729 MHz userspace
15a50000.ofa 780 MHz 115 MHz 780 MHz userspace
17000000.gpu 306 MHz 306 MHz 1.17 GHz nvhost_podgov
bwmgr 2.13 GHz 204 MHz 3.20 GHz bpmp-bwmgr
POWER RAILS
VDD_IN [||] 336/2128 mA
VDD_CPU_GPU_CV [.] 56/32760 mA
VDD_SOC [.] 88/32760 mA
FANS
PWM pwm1 (pwmfan) [||||] 93/255 (36.5%) mode: userspace
TACH rpm (pwm_tach) 2033 RPM
NVPMODEL ACTIVE CAPS
Active: [0] MAXN_SUPER configuration: /etc/nvpmodel.conf
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit (c) 2026, RB


Oct 13 15:23
MAXN SUPER
jtop MAXN SUPER|CPU 8.7%|GPU 0.0%
jtop 7.2.2 - (c) 2026, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats
Platform
Machine: aarch64
System: Linux
Distribution: Ubuntu 24.04 Noble Numbat
Release: 6.8.12-1021-tegra
Python: 3.12.3
Libraries
CUDA: 13.2.86
cuDNN: 9.25.1
TensorRT: 11.2.1.2
VPI: 4.1.4.0
Vulkan: 1.4.321
OpenCV: 4.8.0 with CUDA: NO
CPU
Model: Arm Cortex-A78AE
Architecture: Armv8.2-A
Configuration: 8 cores
Revision: r0p1
MIDR: 0x410fd421
Vector ISA: NEON
Numeric formats: FP16
Integer ML: INT8 Dot Product
Atomics: Arm LSE
Security: PAC
Cryptography: AES, SHA-1, SHA-2, CRC32
Serial Number: [s|XX CLICK TO READ XXX]
Hardware
Model: NVIDIA Jetson Orin NX Engineering Refe
699-level Part Number: 699-13767-0000-303 F.1
P-Number: p3767-0000
Module: NVIDIA Jetson Orin NX (16GB ram)
SoC: tegra234
CUDA Arch BIN: 8.7
L4T: 39.2.1
Jetpack: 7.2.1
Hostname: jetson
Interfaces
enP8p1s0: 192.168.2.55
wlP1p1s0: 192.168.18.23
l4tbr0: 192.168.55.1
docker0: 172.17.0.1
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit
(c) 2026, RB